

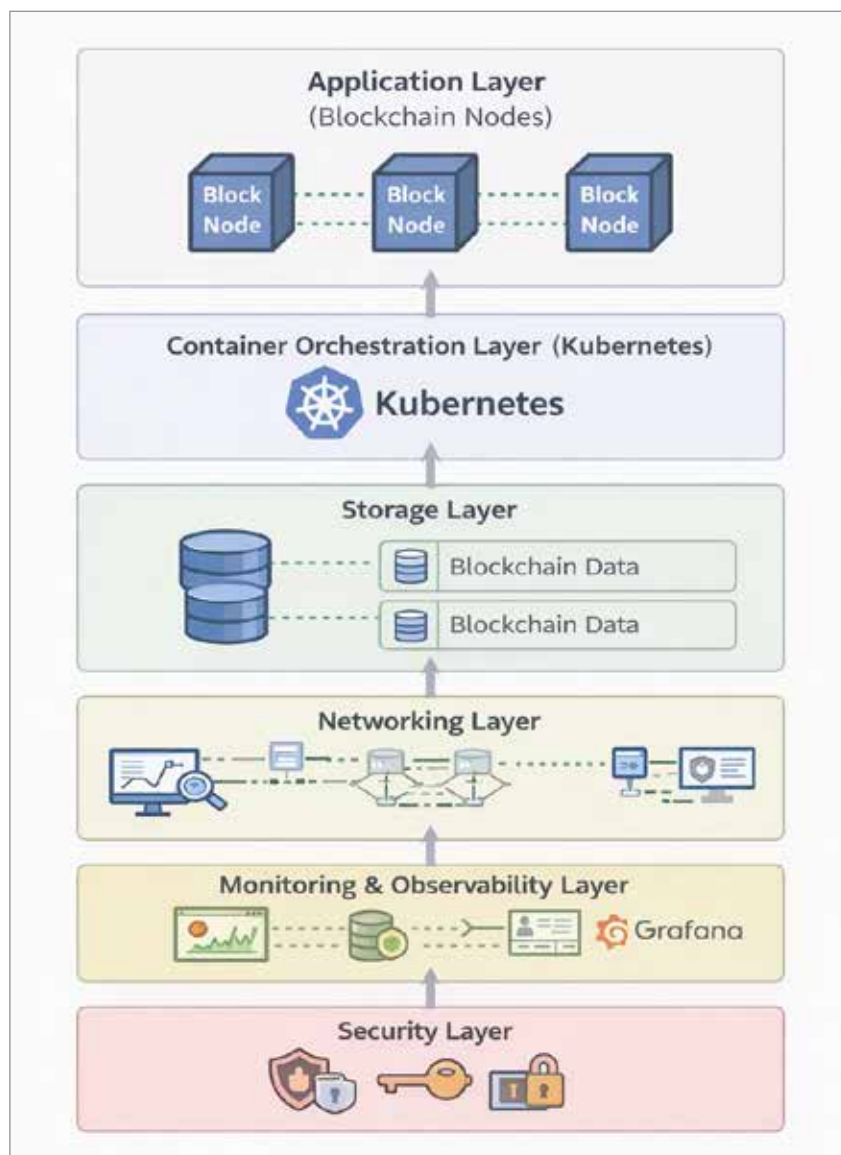


Figure 3: High-level architecture of blockchain nodes running on Kubernetes

of running these nodes on separate virtual machines, they can be deployed as containers within a Kubernetes cluster. Each node can be packaged using a container image such as Geth. Then these nodes can be deployed using StatefulSets, which ensure that each node has a stable identity and persistent storage. Persistent volumes are attached to each node to store blockchain data. This ensures that even if a container is restarted, the node doesn't lose its synchronisation state.

Kubernetes services are used to enable communication between nodes and external applications. This allows nodes to interact with each other and remain connected to the blockchain network. If a node fails, Kubernetes automatically restarts it. Additional nodes can be added easily when required, helping improve availability and distribute load.

This approach simplifies management and ensures that blockchain nodes run reliably without requiring constant manual intervention.

Running blockchain nodes in production is not just about deploying the software; it requires managing a distributed system that demands reliability, scalability, and continuous operation. As blockchain adoption grows, the limitations of traditional infrastructure become more visible. Managing nodes manually using standalone systems does not scale well and introduces operational challenges. Kubernetes offers a practical way to address these challenges. With its capabilities around automation, scaling, and self-healing, it provides a stable foundation for running blockchain node infrastructure.

By combining blockchain technologies with cloud-native platforms like Kubernetes, organisations can build systems that are not only decentralised but also resilient and easier to manage. **END** 🐧

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